Lecture 6.1: From Question to Estimating Equation

AREC 570 / ECON 530 · Unit 6

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Before class

Assigned

- Angrist and Pischke, *Mostly Harmless Econometrics*, Chapters 1 to 3 (Angrist and Pischke 2009)
- Jurajda, “Lecture Notes on Identification Strategies” (Jurajda 2007)
- Timmins and Schlenker, “Reduced-Form Versus Structural Modeling in Environmental and Resource Economics” (Timmins and Schlenker 2009)

Readings and course handouts are available on Canvas.

Also used in these slides

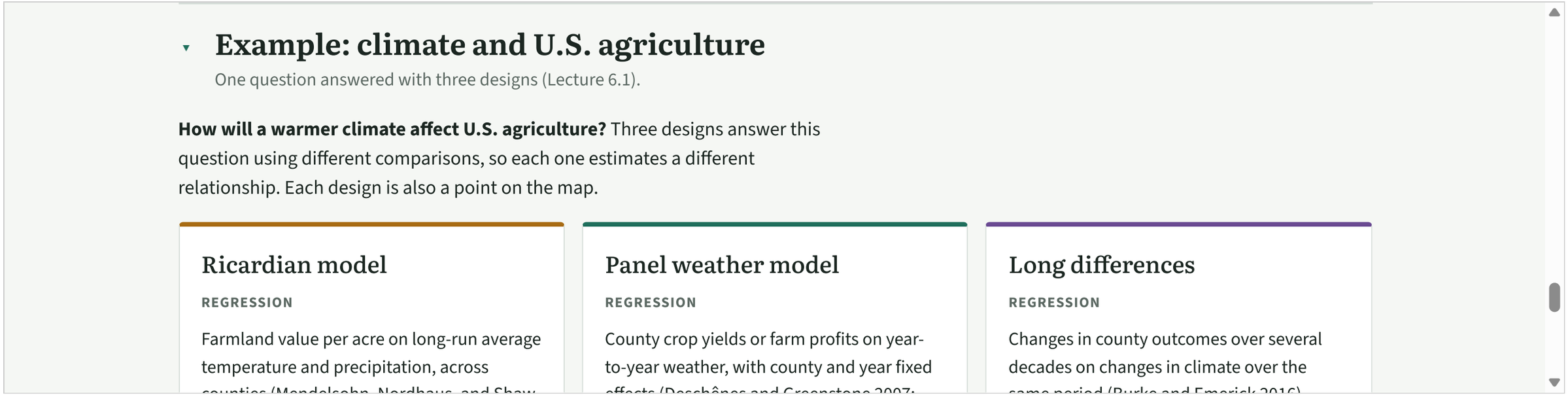
- Bellemare, “How to Write Applied Papers in Economics,” from Lecture 2.3 (Bellemare 2020)
- AJAE Author Guidelines (Agricultural and Applied Economics Association, n.d.)
- Athey and Imbens, “The State of Applied Econometrics,” on robustness and sensitivity (assigned for 6.3) (Athey and Imbens 2017)
- Gentzkow and Shapiro on keys, from Lecture 5.4 (Gentzkow and Shapiro 2014)
- Published example: Dalheimer and Bellemare (2026)
- Veterans example: Angrist (1990); Angrist (1998)
- Climate example: Mendelsohn et al. (1994); Schlenker et al. (2005); Deschênes and Greenstone (2007); Schlenker and Roberts (2009); Burke and Emerick (2016)

Use the empirical strategy checklist to guide what this section should report

- One page checklist, parallel to the data checklist from Lecture 5.5
- Rows 1 to 9 include what every paper should report; rows 10 to 17 depend on the design
- Each row is tagged with the lecture that covers it; today covers rows 1 to 9
- Built from Bellemare's Section 5, Angrist and Pischke's Chapter 1, and the AJAE Author Guidelines

Open the empirical strategy checklist

The methods map helps you match a method to your question



A map of methods, a decision tree, the four choices every empirical strategy states, structural estimation details for 6.2, and the climate example we use today. Use it as a reference while writing.

Open the methods map in a new tab

What must the empirical strategy section accomplish?

Research FAQs

Social science is interested in cause-and-effect relationships

Before starting your research, ask:

- What is the causal relationship of interest?
- What experiment could ideally be used to capture the causal effect of interest?
- What is your identification strategy?
- What is your mode of statistical inference?

These questions are answered throughout the paper, but formally written up in the methods (or empirical strategy) section

Source: Angrist and Pischke (2009), Chapter 1.

Where the empirical strategy section lives in your paper

1. Introduction
2. Background or Literature Review
3. Conceptual Framework
4. Data and Summary Evidence
5. **Empirical Strategy** & **Identification Strategy**
6. Results
7. Discussion (or Policy Implications, or Managerial Implications)
8. Conclusion

Bellemare calls this section the Empirical Framework, and some papers place it before the data section (Bellemare 2020).

The empirical strategy section turns a prediction from your framework into an estimating equation

- Conceptual framework (Unit 4) gives you guidance on developing testable predictions
- Data (Unit 5) gives you guidance on describing your data, the unit of observation, and the measures that make your data fit for use
- Methods (this unit) explains how to turn the conceptual model into an equation with estimable parameters, and why the data identify them

Four terms come up in every empirical strategy section

- **Parameter of interest:** the coefficient that answers your research question
- **Estimator:** the method that turns data into an estimate of that parameter, such as OLS, maximum likelihood, or a difference in means
- **Inference:** how you measure the uncertainty in the estimate, through standard errors, clustering, and hypothesis tests
- **Identifying assumption:** the condition under which the parameter you estimate equals the relationship you want to learn about

Bellemare splits the section into what you estimate and why it identifies a causal effect

Estimation strategy

The equations, the estimator, and how inference is done

Identification strategy

The feature of the data that supports a causal interpretation, or, the manner in which a researcher uses observational data to approximate a real experiment

Use these as the two parts of your section (Bellemare 2020).

Example: Estimation strategy

2.1 Estimation Strategy

We estimate the relationship between the extent of participation in GAVCs by a country in a given year and (i) food price levels as well as (ii) volatility in the same country-year. To do so, we estimate the following baseline equation:

$$\Delta p_{it} = \beta_1 \Delta GAVC_{it} + \gamma_1' \Delta X_{it} + \eta_{1t} + e_{1it}, \quad (1)$$

where p_{it} is the real consumer price level for food in county i in year t, $GAVC_{it}$ is the extent of GAVC participation by the same country in the same year, and X_{it} is a vector of control variables that includes the time-variant country-level characteristics listed in Appendix Table 5. We also include year fixed effects η_t to control for shocks affecting all countries in each given year. Lastly, e_{it} is an error term with mean zero.

In this case, the parameter of interest is β_1 which, in Equation 1, captures the association between participation in GAVCs and the real food price level. We estimate Equation 1

Source: Dalheimer and Bellemare (2026), Section 2.1 (accepted version posted on Bellemare's website).

Example: Identification strategy

2.2 Identification Strategy

When analyzing the impact of trade on food prices, endogeneity poses a significant challenge. One issue could be reverse causation, where changes in food prices influence trade patterns instead of being solely driven by them. For instance, rising food prices in a country might lead to higher imports to meet demand, creating a feedback loop between trade and prices that complicates causal inference.

Omitted variables present another problem, as external factors like extreme weather events, economic policies, or political instability can simultaneously affect both trade volumes and food prices. For example, a natural disaster in an exporting region may disrupt trade while also driving up global prices. Ignoring such variables could result in misleading estimates of the trade-price relationship.

Finally, data limitations and structural market features can introduce additional biases. Inaccuracies in trade or price data, along with complexities such as market concentration

Source: Dalheimer and Bellemare (2026), Section 2.2 (accepted version posted on Bellemare's website).

The following questions help organize the section, typically answered in the order they appear in a paper

Subsection	Question	Where the answer appears
Estimation strategy	1. What is the relationship of interest?	The estimating equation and the parameter of interest
	2. How will you measure uncertainty?	The estimator, the standard errors, the clustering level, and why
Identification strategy	3. Is that relationship causal?	Whether the parameter is causal, and if not, what it describes
	4. What variation identifies it, and what must be true of that variation?	The identifying assumption
	5. What could make the identifying assumption fail?	The threats, and the design choice or robustness check that handles each

Adapted from Angrist and Pischke (2009), Chapter 1, and Bellemare (2020), Section 5.

What’s included in the estimation strategy subsection

Write the equation so a referee never has to ask what a symbol means

- Latin letters for variables (a, b, c, …), Greek letters (α, β, γ, …) for parameters
- Subscripts run from the smallest unit to the largest
- Define every term directly below the equation
- Give each specification its own coefficient labels
- The unit of observation matches the key in your data section

Source: Bellemare (2020), Section 5.1.

The unit of observation describes a row; the key identifies it in the file

The unit tells the reader what each row means. The key lets you check that there is exactly one row per unit before you estimate (Gentzkow and Shapiro 2014).

	Unit of observation	Key
What it is	What one row represents, in words	The field or fields whose values are unique for every row
Panel weather model	A county in a year	County FIPS code and year
Price-gouging data (Lecture 5.5)	One product at one store in one week	Store, product, and week identifiers
Where it appears	The data section and the equation's subscripts	The code that builds and merges the data

The estimation strategy also names the estimator and says how inference is done

- The estimator by name, such as OLS, a linear probability model, probit, or maximum likelihood, and why
- The hypothesis being tested
- What the standard errors are robust to, such as heteroskedasticity
- Whether and how standard errors are clustered, at what level, and why
- Whether sampling weights are used, and how they were constructed

Source: Bellemare (2020), Section 5.1.

Note: Hypothesis testing and identification answer different questions

- **Hypothesis test:** is β different from zero? This depends on the estimator and the standard errors
- **Identification:** does β measure the causal effect? This depends on the identifying assumption
- A significant β is evidence of a causal effect only if β is identified
- More data makes the test more precise, but it does not change what β measures; more data -> smaller standard error -> more precise

The identification strategy subsection explains why β measures the effect you want.

What’s included in the identification strategy subsection

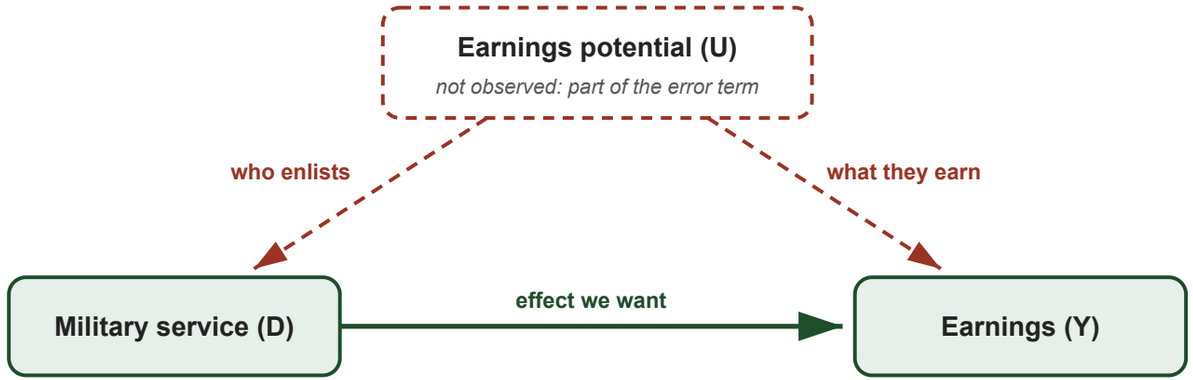
Identification asks whether your data and assumptions determine the parameter you want to estimate

- Suppose you had data on the entire population. The parameter is identified if only one value of it is consistent with those data
- If more than one value fits, the parameter is not identified, and a larger sample or a different estimator will not fix it
- Whether a parameter is identified depends on assumptions about how the data were generated

A directed acyclic graph (DAG) draws those assumptions: each arrow is a causal effect, and each missing arrow is an assumption

- **Example:** a paper wants to estimate the effect of military service on later earnings (Angrist 1990; posed in Jurajda 2007, Example 1.1)

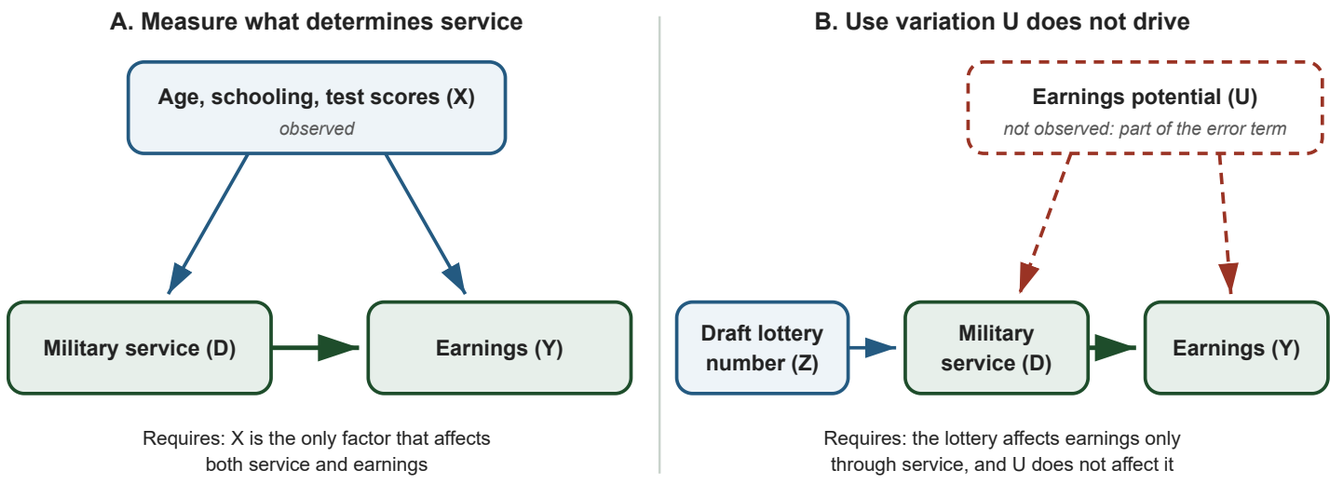
When an unobserved factor drives both treatment and outcome, the effect is not identified



A comparison of veterans and nonveterans picks up both the effect and the path through earnings potential. With unlimited data it still cannot separate them (Angrist 1990).

In regression terms, military service is **endogenous**: earnings potential is part of the error term and also affects who serves, so service is correlated with the error term.

To show one thing causes another (the effect is identified) you must either control for shared causes, or isolate an exogenous source of variation



Both remove the endogeneity. A takes the shared causes out of the error term by controlling for them. B uses only the variation in service that the lottery creates, which is **exogenous**: unrelated to the error term.

A: Angrist (1998), discussed in Angrist and Pischke (2009), Chapter 3. B: the Vietnam-era draft lottery, Angrist (1990), discussed in Jurajda (2007), Example 3.11.

The identifying assumption says why the groups you compare are comparable

- Comparing treated and untreated units measures the treatment effect plus any difference that existed between the groups before treatment
- Example: veterans may earn more because of military service, or because people with higher earnings potential were more likely to enlist (Angrist 1990)
- That pre-existing difference is selection bias
- The identifying assumption states why selection bias is zero for your comparison
- The coefficient receives a causal interpretation because we successfully control for all mutually determining factors, or exploit variation orthogonal to the error term, ensuring the groups are identical on average except for the treatment itself

Bellemare asks every paper to address four threats to identification

Threat	What it means	Common responses
Reverse causality	The outcome affects the treatment	Measure treatment before the outcome; use an instrument
Unobserved heterogeneity	An omitted factor affects both treatment and outcome	Fixed effects, controls, or a design that holds the factor constant
Measurement error	The treatment is recorded with error; classical error biases the estimate toward zero	A second measure; an instrument
Spillovers	One unit's treatment affects another unit's outcome, now or later	Define units so spillovers stay within them; test for them

Bellemare calls the first three sources of **endogeneity**: each makes the treatment correlated with the error term (Bellemare 2020, sec. 5.2).

Threats to identification and problems with inference are different

- **Problem with inference:** the estimate may measure the right thing, but its standard errors are too small, so the result looks more certain than it is
- **Threat to identification:** the estimate measures the wrong thing, and more data will not fix it
- Clustering standard errors is an inference choice: it changes the standard errors, not the estimate

Source: Bellemare (2020), Sections 5.1 and 5.2.

Robustness check or sensitivity analysis?

Robustness checks vary your choices

- Re-estimate your model with another defensible outcome measure, treatment measure, estimator, functional form, or sample, and show the conclusion holds (Bellemare 2020, sec. 6.2)
- A placebo test estimates an effect that should be zero; Bellemare counts these as robustness checks, and Athey and Imbens list them separately

Sensitivity analyses relax your assumptions

- Purpose is to ask how far the identifying assumption would have to fail before the conclusion changes, for example how strong an omitted variable would have to be (Athey and Imbens 2017)

Why papers include them: researchers try many specifications before choosing one, so readers need evidence that the result was not selected for (Athey and Imbens 2017)

Example: climate and U.S. agriculture

Three published papers answer the same climate question with different empirical strategy sections

Question: How will a warmer climate affect U.S. agriculture?

- The ideal experiment: give some counties a permanently warmer climate, then wait decades while farmers adapt
- No paper can run that experiment, so each one approximates it with a different comparison
- The comparison determines what the coefficient measures
- Read each design as the empirical strategy section its paper would write: estimation strategy, then identification strategy

The Ricardian model compares farmland values across counties with different climates

Estimation strategy

$$V_c = \alpha + f(T_c, P_c) + X_c^{'}\gamma + \varepsilon_c$$

- V_c: farmland value per acre in county c
- T_c, P_c: long-run average temperature and precipitation
- $f(\cdot)$: a quadratic; its slope in T_c is the parameter of interest
- X_c: soil quality and other county characteristics
- Standard errors allow neighboring counties' errors to be correlated

Identification strategy

- **Identifying assumption:** counties with different climates do not differ in other ways that affect land values
- **Main threat:** irrigated farms in drier, hotter counties draw water from elsewhere and have higher land values; Schlenker, Hanemann, and Fisher re-estimate the model for dryland counties (Schlenker et al. 2005)
- **What the coefficient measures:** the long-run effect of climate on land values, including adaptation

Source: Mendelsohn et al. (1994).

The panel model compares each county with itself across years, which removes irrigation but changes what is estimated

Estimation strategy

$$y_{ct} = g(W_{ct}) + \gamma_c + \delta_t + \varepsilon_{ct}$$

- y_{ct}: log crop yield or farm profit in county c, year t
- W_{ct}: growing-season weather, such as degree days and precipitation
- $g(\cdot)$: its slope is the parameter of interest
- γ_c, δ_t: county and year fixed effects
- Standard errors allow neighboring counties' errors to be correlated, because they share weather shocks

Identification strategy

- **Identifying assumption:** within a county, hotter and cooler years do not differ in other ways that affect yields
- **Main threats:** measurement error in gridded weather, and price changes that spill over across counties
- **What the coefficient measures:** the short-run response to weather; farmers cannot fully adapt to one hot year, so the long-run effect can be larger or smaller (Timmins and Schlenker 2009)

Sources: Deschênes and Greenstone (2007); Schlenker and Roberts (2009).

Long differences compare slow changes in climate over decades, which recovers the adaptation the panel model misses

Estimation strategy

$$\Delta\bar{y}_c = \alpha_s + \beta\,\Delta\bar{W}_c + \Delta\bar{\varepsilon}_c$$

- $\Delta\bar{y}_c$: change in county c's multi-year average outcome between two periods
- $\Delta\bar{W}_c$: change in its multi-year average weather over the same periods
- α_s: state fixed effects; β: the parameter of interest
- Standard errors allow neighboring counties' errors to be correlated; one observation per county leaves far fewer observations than the panel

Identification strategy

- **Identifying assumption:** within a state, counties that warmed more did not change in other ways that affect yields
- **Main threat:** other trends, such as technology adoption, that move with climate trends
- **What the coefficient measures:** the response to gradual climate change, including the adaptation farmers made over the period

Source: Burke and Emerick (2016).

The three sections estimate different things, and each states it

	Ricardian	Panel weather	Long differences
Estimation: comparison	Across counties	Within a county, across years	Across counties in a state, by how much each warmed
Estimation: standard errors	Correlated across neighboring counties	Correlated across neighboring counties	Correlated across neighboring counties; fewer observations
Identification: assumption	Counties do not differ in other ways	Hotter and cooler years do not differ in other ways	Counties that warmed more did not change in other ways
Identification: main threat	Irrigation	Measurement error; price spillovers	Other trends, such as technology
What the coefficient measures	Long-run effect, including adaptation	Short-run response to weather	Response to gradual change, including adaptation

Each design has to answer all four threats

Threat	Ricardian	Panel weather	Long differences
Reverse causality	Land values do not change the climate	Farm output does not change the weather	Farm output does not change climate trends
Unobserved heterogeneity	The main threat: irrigation and other factors that vary with climate	County and year fixed effects absorb fixed differences and national shocks	The main threat: other trends that move with climate trends
Measurement error	Reported land values may differ from sale prices	Gridded weather misses conditions on the farm	Averaging over several years reduces noise in weather
Spillovers	Land values reflect current crop prices, which warming everywhere would change	Year effects absorb national price changes	State effects absorb price changes common within a state

Each cell needs an answer: the threat does not apply, the design handles it, or a robustness check tests it.

Writing the section

Put the five questions in your draft as comments, then answer each in prose

```latex
\section{Empirical Strategy}

\subsection{Estimation Strategy}
% What is the relationship of interest?
%   equation, every term defined, parameter of interest
% How will you measure uncertainty?
%   standard errors, clustering level, and why

\subsection{Identification Strategy}
% Is that relationship causal?
% What variation identifies it, and what must be true of that variation?
% What could make the identifying assumption fail?
%   each threat, and the design choice or robustness check that handles it
```

Lines that start with `%` do not appear in the compiled PDF. In Word, use comments; in Quarto or Markdown, use `<!-- -->`.

Writing task: your estimating equation and identifying assumption

For your own proposal:

1. Write the estimating equation
2. Define every term, including subscripts and the unit of observation
3. Name the parameter of interest and say whether it is causal
4. State the identifying assumption in one sentence

References

Agricultural and Applied Economics Association. n.d. *American Journal of Agricultural Economics: Author Guidelines*. Oxford University Press. Accessed October 27, 2026. https://academic.oup.com/ajae/pages/Author_Guidelines.

Angrist, Joshua D. 1990. “Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records.” *American Economic Review* 80 (3): 313–35. https://economics.mit.edu/sites/default/files/publications/Angrist%201990%20-%20Lifetime%20Earnings%20and%20the%20Vietname%20.pdf.

Angrist, Joshua D. 1998. “Estimating the Labor Market Impact of Voluntary Military Service Using Social Security Data on Military Applicants.” *Econometrica* 66 (2): 249–88. https://doi.org/10.2307/2998558.

Angrist, Joshua D., and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist’s Companion*. Princeton University Press.

Athey, Susan, and Guido W. Imbens. 2017. “The State of Applied Econometrics: Causality and Policy Evaluation.” *Journal of Economic Perspectives* 31 (2): 3–32. https://doi.org/10.1257/jep.31.2.3.

Bellemare, Marc F. 2020. *How to Write Applied Papers in Economics*. Working paper, University of Minnesota.

Burke, Marshall, and Kyle Emerick. 2016. “Adaptation to Climate Change: Evidence from US Agriculture.” *American Economic Journal: Economic Policy* 8 (3): 106–40. https://doi.org/10.1257/pol.20130025.

Dalheimer, Bernhard, and Marc F. Bellemare. 2026. “Global Agricultural Value Chains and Food Prices.” *American Journal of Agricultural Economics* 108 (5): 1282–99. https://doi.org/10.1111/ajae.70030.

Deschênes, Olivier, and Michael Greenstone. 2007. “The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather.” *American Economic Review* 97 (1): 354–85. https://doi.org/10.1257/aer.97.1.354.

Deschênes, Olivier, and Michael Greenstone. 2012. “The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather: Reply.” *American Economic Review* 102 (7): 3761–73.

https://doi.org/10.1257/aer.102.7.3761.
Fisher, Anthony C., W. Michael Hanemann, Michael J. Roberts, and Wolfram Schlenker. 2012. “The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather: Comment.” *American Economic Review* 102 (7): 3749–60. https://doi.org/10.1257/aer.102.7.3749.
Gentzkow, Matthew, and Jesse M. Shapiro. 2014. *Code and Data for the Social Sciences: A Practitioner’s Guide*. University of Chicago mimeo. https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf.
Jurajda, Štěpán. 2007. *Lecture Notes on Identification Strategies*. Lecture notes.
Mendelsohn, Robert, William D. Nordhaus, and Daigee Shaw. 1994. “The Impact of Global Warming on Agriculture: A Ricardian Analysis.” *American Economic Review* 84 (4): 753–71.
Schlenker, Wolfram, W. Michael Hanemann, and Anthony C. Fisher. 2005. “Will U.S. Agriculture Really Benefit from Global Warming? Accounting for Irrigation in the Hedonic Approach.” *American Economic Review* 95 (1): 395–406. https://doi.org/10.1257/0002828053828455.
Schlenker, Wolfram, and Michael J. Roberts. 2009. “Nonlinear Temperature Effects Indicate Severe Damages to U.S. Crop Yields Under Climate Change.” *Proceedings of the National Academy of Sciences* 106 (37): 15594–98. https://doi.org/10.1073/pnas.0906865106.
Timmins, Christopher, and Wolfram Schlenker. 2009. “Reduced-Form Versus Structural Modeling in Environmental and Resource Economics.” *Annual Review of Resource Economics* 1: 351–80. https://doi.org/10.1146/annurev.resource.050708.144119.

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